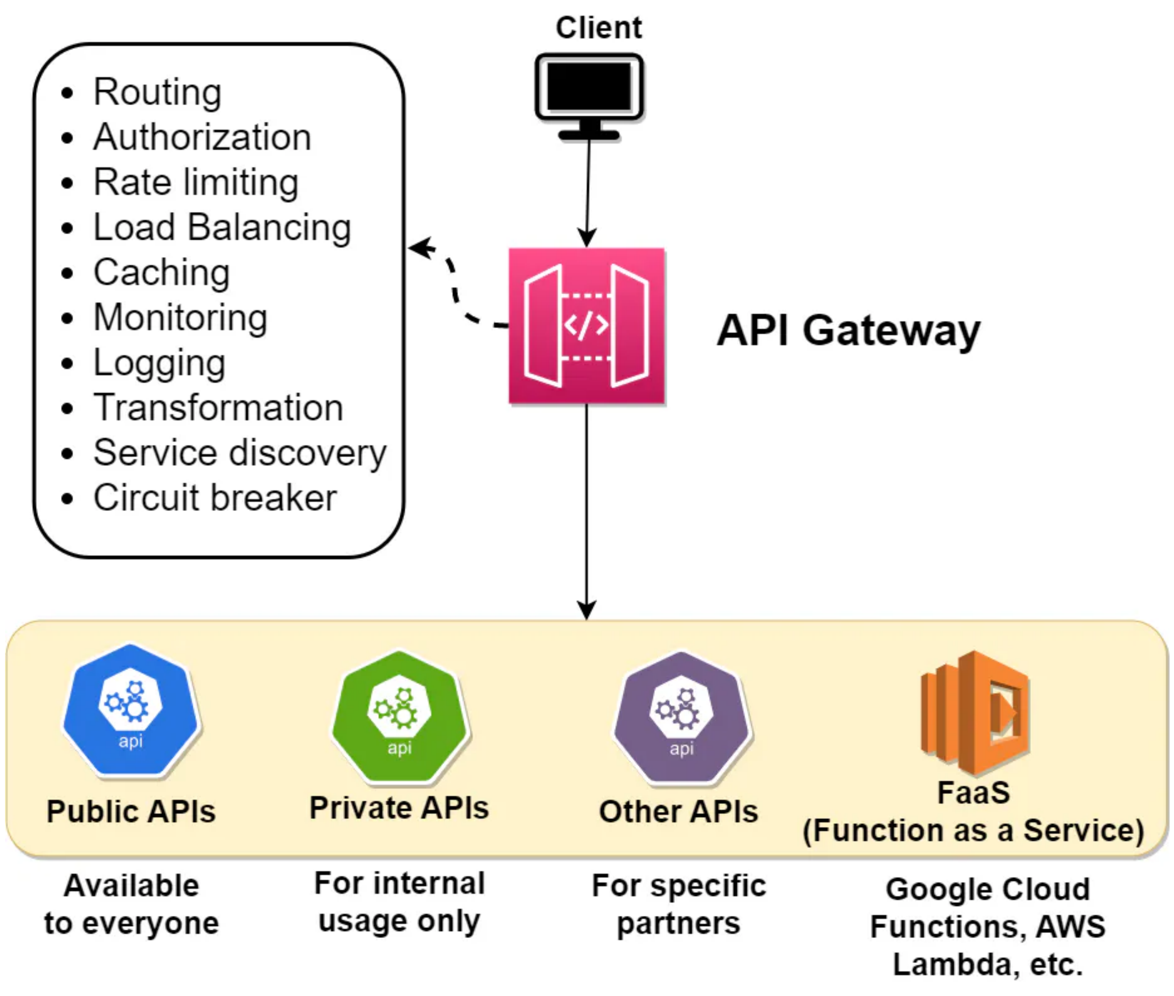
Introduction to API Gateway

An API Gateway is a server-side architectural component in a software system that acts as an intermediary between clients (such as web browsers, mobile apps, or other services) and backend services, microservices, or APIs.

Its main purpose is to provide a single entry point for external consumers to access the services and functionalities of the backend system. It receives client requests, forwards them to the appropriate microservice, and then returns the server's response to the client.

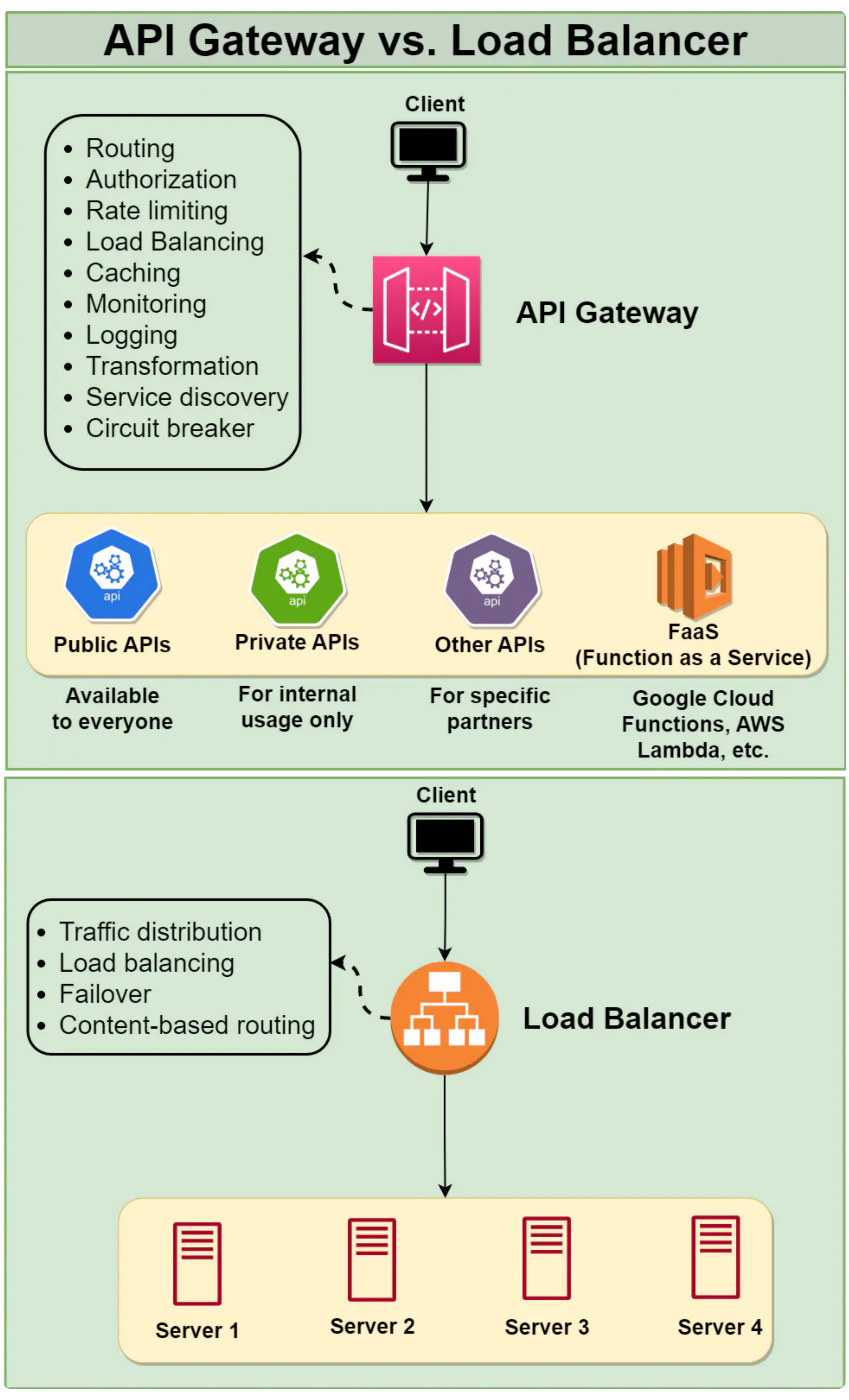
The API gateway is responsible for tasks such as routing, authentication, and rate limiting. This enables microservices to focus on their individual tasks and improves the overall performance and scalability of the system.


Client
Routing
Authorization
Rate limiting
Load Balancing
Caching
Monitoring
Logging
Transformation
Service discovery
Circuit breaker
API Gateway
Public APIs
Private APIs
Other APIs
FaaS
(Function as a Service)
Available to everyone
For internal usage only
For specific partners
Google Cloud Functions, AWS Lambda, etc.

Difference between an API gateway and a load balancer

An API gateway is focused on **routing** requests to the appropriate microservice, while a load balancer is focused on **distributing** requests evenly across a group of backend servers.

An API gateway and a load balancer are both types of infrastructure that can be used in a computer network to manage incoming requests and enhance the performance of a system. However, they work in different ways and serve different purposes.


API Gateway vs. Load Balancer
Client
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Client
Traffic distribution
Load balancing
Failover
Content-based routing
Load Balancer
Server 1
Server 2
Server 3
Server 4

API gateway: An API gateway is a type of middleware that sits between a client and a collection of microservices. Its main purpose is to route requests from clients to the appropriate microservice and then to return the response from the microservice back to the client. An API gateway can also perform other tasks such as authorization, rate limiting, and caching.

Load balancer: A load balancer, on the other hand, is a type of infrastructure that distributes incoming requests evenly across a group of backend servers in order to improve the performance and availability of a system. Load balancers are typically used to handle requests that are sent to a single, well-known IP address, and then route them to one of many possible backend servers based on factors such as server performance and availability.

Another difference between the two is the type of requests that they typically handle. An API gateway is typically used to handle requests for APIs, which are web-based interfaces that allow applications to interact with each other over the internet. These requests typically have a specific URL that identifies the API that the client is trying to access, and the API gateway routes the request to the appropriate microservice based on this URL. A load balancer, on the other hand, is typically used to handle requests that are sent to a single, well-known IP address, and then routes them to one of many possible backend servers based on factors such as server performance and availability.

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